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Distributed variational bayesian meta-learning over multi-agent systems ⋆

Xianyang Zhang [a], Weijie Kang [a,*], Ying Li [b], Bing He [a], Honglan Huang [a]

[a] PLA Rocket Force University of Engineering, Tongxin Road, Baqiao District, Xi'an, Shaanxi, 710025, China
[b] Medical Experiment Center, China Academy of Chinese Medical Sciences, No. 16 Dongzhimennei Nanxiao Street, Dongcheng District, Beijing, 100700, China

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ABSTRACT

Learning new tasks from limited data inherently introduces uncertainty in meta-learning. While Bayesian meta-learning aims to enhance traditional algorithms by quantifying uncertainty in model parameters, existing methods are centralized and cannot handle decentralized data. In this paper, we present the Distributed Variational Bayesian Meta-Learning (DVBML) framework, the first to enable distributed uncertainty quantification for decentralized meta-learning. Unlike deterministic approaches like Dif-MAML, DVBML models meta-parameters as probability distributions. We propose a novel gradient-diffusion hybrid algorithm that combines local stochastic gradient variational Bayes steps with a global diffusion-based consensus strategy, allowing agents to collaboratively learn and propagate both the mean and variance of task parameters. Rigorous theoretical analysis proves that this strategy guarantees convergence and achieves a linear speedup $\mathcal{O}(1/\sqrt{NT})$ in variance reduction compared to non-cooperative baselines. Extensive evaluations on few-shot regression and classification benchmarks show that DVBML matches the accuracy of centralized Bayesian meta-learning and delivers reliable, well-calibrated uncertainty estimates. It significantly outperforms non-cooperative baselines and quantitatively surpasses deterministic distributed methods in calibration metrics, maintaining remarkable robustness even under imbalanced data settings.

1. Introduction

Although machine learning, particularly deep learning, has made remarkable advancements in various domains such as speech and image understanding in recent years (Hinton et al., 2012; Krizhevsky et al., 2017; Nguyen et al., 2020), these methods often rely on a large number of training samples to achieve significant notable performance. In contrast, human can quickly master new learning tasks with only a few training examples. This ability stems from their capability to leverage prior knowledge, thus accelerating learning on novel tasks (Schmidt, 2009), while machine learning algorithms typically start from scratch when faced with a new task. Meta-learning, or "learning to learn", aims to enhance generalization in scenarios with limited samples by leveraging prior knowledge acquired from related tasks. This paper considers a distributed scenario where the databases are scattered over a multi-agent system, and sharing data between different agents is not feasible. Learning from local data is insufficient due to the lack of diversity in the dataset. With the development of multi-agent systems, the observed databases become more complex, therefore, there is a growing need to develop more scalable and efficient distributed meta-learning methods.

Learning new tasks from a few examples might inherently introduce uncertainty, even with the best possible prior. This uncertainty becomes even more pronounced over multi-agent systems, where each agent can only observe a small portion of the data. It is crucial to accurately quantify this uncertainty, especially in practical applications such as Bayesian optimization (Shahriari et al., 2016), model-based reinforcement learning (Chua et al., 2018), and in safety-critical scenarios. Bayesian inference provides an elegant framework to reason about the uncertainty. To address the problem of uncertainty in meta-learning, several Bayesian meta-learning (BML) approaches have been proposed recently, which reinterpret meta-learning from a Bayesian hierarchical modeling perspective (Amit & Meir, 2018; Finn et al., 2018; Gordon et al., 2018; Grant et al., 2018; Ravi & Beatson, 2018; Yoon et al., 2018). However, these BML methods are centralized algorithms that require all data to be collected on a central agent. In a distributed scenario where datasets are spread across a multi-agent system and cannot be collected in a central node, standard BML approaches become ineffective. The challenge posed by the distributed scenario has motivated us to design a distributed BML method that can adapt to such circumstances, where datasets are decentralized and stored by different agents over a multi-agent system.

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* Corresponding author.
E-mail addresses: zhangxyang177@163.com (X. Zhang), milankang@foxmail.com (W. Kang), liying1251@163.com (Y. Li), hebing202301@163.com (B. He), huanghonglan17@alumni.nudt.edu.cn (H. Huang).

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A common ground to address distributed machine learning problems is to formulate them as distributed optimization problems, which can then be solved by various distributed algorithms (Cattivelli & Sayed, 2009; Chen et al., 2021; Dimakis et al., 2010; Takahashi et al., 2010; Vlaski & Sayed, 2021). However, the literatures suggest only a few methods or architectures that are appropriate to distributed meta-learning (Kayaalp et al., 2022; Zhang et al., 2022). For example, in Kayaalp et al. (2022), a diffusion-based model-agnostic meta-learning (Dif-MAML) approach was proposed. Dif-MAML builds upon the MAML framework (Finn et al., 2017) and incorporates a diffusion strategy (Sayed, 2014). However, this method is essentially a distributed version of MAML, and since MAML itself lacks the ability to reason about uncertainty, Dif-MAML is unable to quantify the uncertainty of the model. By considering these limitations and challenges, our goal is to develop a distributed meta-learning algorithm that can effectively reason about uncertainty.

It is also worth noting a parallel line of work on federated Bayesian meta-learning or variational federated learning (Jeon et al., 2024; Liu et al., 2023; Tahir et al., 2024; Wu & Zhang, 2025), which introduces Bayesian reasoning within a centralized client-server architecture to address data heterogeneity across clients. While these methods excel at personalization under server coordination, they are architecturally distinct from our focus on fully decentralized multi-agent systems. Crucially, federated learning relies on a central server, whose failure would halt the entire process. Our target scenario, in contrast, involves agents that communicate solely with neighbors in a peer-to-peer network, demanding solutions that are robust to single-point failures and inherently decentralized.

By considering these limitations and challenges, we introduce the first distributed Bayesian meta-learning framework, named Distributed Variational Bayesian Meta-Learning (DVBML). Our approach fundamentally differs from deterministic distributed methods like Dif-MAML (Kayaalp et al., 2022) by models meta-parameters as probability distributions, enabling explicit uncertainty quantification through variational posteriors. Our approach departs from deterministic methods like Dif-MAML by modeling meta-parameters as probability distributions, enabling explicit uncertainty quantification in model weights through variational posteriors.

While our framework builds upon established pillars—variational inference for Bayesian learning and diffusion strategies for decentralized consensus—its novelty lies in the original synthesis of these components to solve a new problem: distributed, uncertainty-aware few-shot meta-learning. This synthesis allows each agent to locally reason about parameter uncertainty while collaboratively constructing a robust global prior through the exchange of statistical summaries, directly overcoming the limitations of point-estimation in decentralized settings. The main contributions of this paper are summarized as follows.

1. We establish Distributed Variational Bayesian Meta-Learning (DVBML), a novel framework to enable distributed uncertainty quantification in model parameters for decentralized few-shot learning. By modeling meta-parameters as probability distributions and leveraging variational inference, each agent locally approximates task-specific posterior distributions while exchanging statistical summaries via a diffusion strategy. This effectively addresses the limitation of deterministic methods, such as Dif-MAML, which lack uncertainty reasoning capabilities.
2. Under this framework, we propose a novel gradient-diffusion hybrid algorithm that integrates local stochastic gradient variational Bayes steps with a global diffusion-based "adapt-then-combine" strategy. This integration is key, as it explicitly models and propagates both the mean and variance of task-specific parameters, allowing all agents to converge toward a consensus posterior distribution that robustly captures task uncertainty.
3. We provide rigorous theoretical guarantees and extensive empirical validation for the proposed algorithm. Mathematically, we prove its convergence to a stationary point and demonstrate a linear speedup $\mathcal{O}(1/\sqrt{NT})$ in variance reduction. Empirically, extensive evaluations on few-shot benchmarks demonstrate that DVBML not only matches the accuracy of its centralized Bayesian counterpart but also delivers reliable, well-calibrated uncertainty estimates—significantly outperforming non-cooperative baselines and quantitatively surpassing deterministic distributed methods in calibration metrics (ECE/MCE), even under imbalanced data partitions.

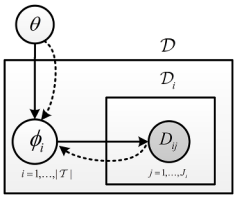


Fig. 1. Hierarchical graphical models for variational bayesian meta learning. Dotted and solid lines denote the variational approximations and the generative model, respectively. Note that a prior parameterised by θ is shared across many tasks.

The remainder of this paper is organized as follows. Section 2 formulate the distributed Bayesian meta-learning problem. Section 3 establishes a Distributed Variational Bayesian meta-learning framework and deduces the gradient-based DVBML algorithm. Section 4 experimentally demonstrates the effectiveness of the proposed methods. Finally, Section 5 concludes this paper and further discusses promising avenues for future work.

2. Problem formulation

2.1. Traditional variational Bayesian meta-Learning

In traditional variational Bayesian meta-learning (VBML), which is a centralized approach, we consider an observed task set $\mathcal{T} = \{\tau_i\}_{i=1}^{|\mathcal{T}|}$ consisting of $|\mathcal{T}|$ task. Each individual task τ_i contains a data set $\mathcal{D}_i = \{D_{ij}\}_{j=1}^{J_i}$. The core idea of meta-learning involves acquiring a meta model (or meta-knowledge) on task sets $\mathcal{T}$, so that when given any specific task, a few training samples in the task is sufficient to obtain a well-generalized model. From Bayesian perspective, the meta model acts as a suitable prior, enabling the derivation of an effective posterior with the help of the likelihood on the task-specific data.

Considering θ as the parameter of the meta-level and ϕ_i as the task-specific parameters for task τ_i. Then we can view θ and ϕ_i as the global and local variables (parameters), respectively. The global variables θ is used to infer the posterior distribution of ϕ_i. Fig. 1 shows the hierarchical graph model of BML.

In Bayesian meta-learning, the objective is to minimize the negative evidence lower bound (ELBO) (Ravi & Beatson, 2018):

$$\arg\min_{\psi} \mathcal{L}(\psi) = \mathbb{E}_{q(\theta;\psi)}\left[\sum_{i=1}^{|\mathcal{T}|} -\mathbb{E}_{q(\phi_i;\lambda_i)}[\log p(\mathcal{D}_i|\phi_i)] + \mathbb{KL}[q(\phi_i;\lambda_i)||p(\phi_i|\theta)]\right] + \mathbb{KL}[q(\theta;\psi)||p(\theta)], \tag{1}$$

where $p(\theta)$ is the prior of the meta parameters θ, and $p(\phi_i|\theta)$ is the conditional prior for the task-specific parameters ϕ_i. $q(\theta;\psi)$ and $q(\phi_i;\lambda_i)$ are the variational distributions to approximate $p(\theta)$ and $p(\phi_i|\theta)$, respectively. $\mathbb{KL}$ is the Kullback-Leibler divergence between two distributions. Our goal is to find the optimal variational distribution $q(\theta^*;\psi^*)$ of the problem (1).

In few-shot meta-learning, a common method to solve the problem (1) is to employ bi-level optimization, which involves inner update and outer update. We split the dataset $\mathcal{D}_i$ for each task into support set $\mathcal{D}_i^s$ and query set $\mathcal{D}_i^q$, i.e., $\mathcal{D}_i = \mathcal{D}_i^s \bigcup \mathcal{D}_i^q$. The inner update aims to derive optimal task-specific parameters λ_i^* for the task τ_i by optimize the

following local evidence lower bound,

$$\arg\min_{\lambda_i} L(\lambda_i; \mathcal{D}_i^s) \triangleq \mathbb{KL}[q(\phi_i;\lambda_i)||p(\phi_i|\theta)] - \mathbb{E}_{q(\phi_i;\lambda_i)}[\log p(\mathcal{D}_i^s|\phi_i)]. \tag{2}$$

Usually, only K steps are performed in the inner update with a fixed θ, i.e.,

(1) $\lambda_i^{(0)} = \psi$;

(2) **For** $k = 1, \cdots, K$:

$$\lambda_i^{(k)} = \lambda_i^{(k-1)} - \beta_{in}\nabla_{\lambda_i} L(\lambda_i^{(k-1)}; \mathcal{D}_i^s); \tag{3}$$

End For

(3) $\lambda_i^* = \lambda_i^{(K)}$.

where β_{in} is the inner learning rate. $\lambda_i^{(0)} = \psi$ is the initialization for the task τ_i. The outer update is to optimize the meta-objective $\mathcal{L}(\psi)$ on the query set $\{\mathcal{D}_i^q\}_{i=1}^{|\mathcal{T}|}$ by gradient descent,

$$\psi(t+1) = \psi(t) - \beta_o \nabla_\psi \mathcal{L}(\psi(t)), \tag{4}$$

where β_o is the outer learning rate, and

$$\mathcal{L}(\psi) = \mathbb{E}_{q(\theta;\psi)}[\sum_{i=1}^{|\mathcal{T}|} L(\lambda_i^{(K)}; \mathcal{D}_i^q)] + \mathbb{KL}[q(\theta;\psi)||p(\theta)].$$

The inner update in Eq. (3) aims to fine-tune the task-specific parameters for a given task, while the outer update in Eq. (4) optimizes the meta-parameters across all the sampled tasks. It is worth noting that the outer update need the entire task set $\mathcal{T}$ to update parameters. However, in distributed scenarios involving multi-agent systems, each individual agent can only access a subset of the entire task set, which makes the traditional variational Bayesian meta-learning fail to work. Our aim is to develop a distributed form of variational Bayesian meta-learning algorithm over multi-agent system, where each agent can only access to its own task set.

2.2. Distributed meta-learning formulation

In distributed meta-learning settings, we consider a multi-agent system consisting of N agents that communicate over an undirected network represented by a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$, which consists of the set of agents or nodes $\mathcal{V} = \{1, 2, \ldots, N\}$, the set of edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$, and the weighted adjacent matrix $\mathcal{A} = [a_{nm}]$, where a_{nm} denotes the matrix entry in the n-th row and m-th column. In the weighted adjacent matrix $\mathcal{A}$, all elements are nonnegative, row stochastic and the diagonal elements are all positive, i.e., $a_{nn}0, a_{nm} \geq 0, \sum_{m\in\mathcal{V}} a_{nm} = 1$. If $a_{nm}0, m \neq n$, there is an edge $(n, m) \in \mathcal{E}$, which means node n can directly receive the information of node m, and node m is called the neighbor of node n. All neighbors of node n can be represented by the set $\{m \in \mathcal{V}|(n, m) \in \mathcal{E}\} \triangleq \mathcal{N}_n^0$. Also, $\mathcal{N}_n^0 \bigcup \{n\} \triangleq \mathcal{N}_n$. $\mathcal{G}$ is called strongly connected if for any pair nodes (i_1, i_l), there exists a directed path from n_1 to n_l consisting of edges $(i_1, i_2), (i_2, i_3), \ldots, (i_{l-1}, i_l)$, where $n_2, \ldots, i_{l-1}$ are nodes. An undirected graph $\mathcal{G}$ is simply called connected if it is strongly connected. Let $|\cdot|$ denote the cardinality of a set.

In this section, we reformulate the optimization problem (1) in distributed scenario, where the total task set $\mathcal{T}$ is distributed and stored across a multi-agent system without sharing it. We use the subscript n to represent the quantity associated with agent n. For example, $\mathcal{T}_n = \{\tau_{n,i}\}_{i=1}^{|\mathcal{T}_n|}$ denotes the task set managed by agent n, and $\mathcal{D}_{n,i}$ is the data set related to task $\tau_{n,i}$ observed by agent n.

As mentioned above, each agent cannot access the entire observed data set $\mathcal{T}$ in distributed setting, which makes the distributed meta-leaning problem difficult. Fortunately, the meta-objective function $\mathcal{L}(\psi)$ can be decomposed into the sum of several functions, each of which is only associated to the local data, as shown below:

$$\begin{aligned}
\mathcal{L}(\psi) &= \mathbb{E}_{q(\theta;\psi)}\left[\sum_{i=1}^{|\mathcal{T}|} L(\lambda_i^{(K)}; \mathcal{D}_i^q)\right] + \mathbb{KL}[q(\theta;\psi)||p(\theta)] \\
&= \mathbb{E}_{q(\theta;\psi)}\left[\sum_{n=1}^{N}\sum_{i=1}^{|\mathcal{T}_n|} L(\lambda_i^{(K)}; \mathcal{D}_{n,i}^q)\right] + \mathbb{KL}[q(\theta;\psi)||p(\theta)] \\
&= \sum_{n=1}^{N} \mathbb{E}_{q(\theta;\psi)}\left[\sum_{i=1}^{|\mathcal{T}_n|} L(\lambda_i^{(K)}; \mathcal{D}_{n,i}^q)\right] + \sum_{n=1}^{N}\frac{1}{N}\mathbb{KL}[q(\theta;\psi)||p(\theta)] \\
&= \sum_{n=1}^{N}\left[\mathbb{E}_{q(\theta;\psi)}[\sum_{i=1}^{|\mathcal{T}_n|} L(\lambda_i^{(K)}; \mathcal{D}_{n,i}^q)] + \frac{1}{N}\mathbb{KL}[q(\theta;\psi)||p(\theta)]\right] \\
&= \sum_{n=1}^{N} \mathcal{L}_n(\psi),
\end{aligned} \tag{5}$$

where

$$\mathcal{L}_n(\psi) \triangleq \mathbb{E}_{q(\theta;\psi)}[\sum_{i=1}^{|\mathcal{T}_n|} L(\lambda_i^{(K)}; \mathcal{D}_{n,i}^q)] + \frac{1}{N}\mathbb{KL}[q(\theta;\psi)||p(\theta)].$$

Note that $\mathcal{L}_n(\psi)$ only relies on the local task set $\mathcal{T}_n$ obtained by agent n. Then the optimization problem (1) can be rewritten as

$$\arg\min_{\psi} \sum_{n=1}^{N} \mathcal{L}_n(\psi). \tag{6}$$

Problem (6) illustrates that the global meta-objective can be decomposed into the aggregation of local objectives. Despite the decomposition of the global meta-objective $\mathcal{L}(\psi)$, it is important to note that individually minimizing the local objective $\mathcal{L}_n(\psi)$ at each agent doesn't necessarily yield the global optimal solution (Hua & Li, 2015). Our work is to establish a distributed framework for variational Bayesian meta-learning so as to solve problem (6).

3. Proposed algorithm

In this section, we first construct a Distributed Variational Bayesian Meta-Learning (DVBML) framework that combines the ideas of distributed optimization methods and a diffusion strategy, then propose a gradient descent-based DVBML algorithm under Gaussian parameterization.

3.1. Distributed variational Bayesian meta-learning framework

Similar to the conventional (centralized) variational Bayesian meta-learning algorithm, DVBML also consists of two updates: inner update and outer update. As shown in Eq. (3), the inner update perform multiple steps of gradient descent on each individual tasks, so given the global parameters ψ_n. The inner update can be performed individually at each agent:

$$\lambda_{n,i}^{(k+1)} = \lambda_{n,i}^{(k)} - \beta_{in}\nabla_\lambda L(\lambda_{n,i}^{(k)}; \mathcal{D}_{n,i}^s), k = 0, 1, \cdots, K-1. \tag{7}$$

Here $\lambda_{n,i}$ denotes the task-specific variational parameters on the task $\tau_{n,i}$ at agent n, initialized by the global meta-parameters ψ_n and optimized via K inner gradient descent steps (Eq. (7)) to approximate the posterior distribution $q(\phi_i; \lambda_{n,i})$.

In the outer update, to optimize the global objective cooperatively, each agent not only needs to update parameters based on local data, but also needs to share parameter information with neighboring agents. Therefore, we introduce a diffusion-based "adapt-then-combine (ATC)" strategy for the outer update. Instead of directly updating parameters using local data to yield a local optimum at each agent, we utilize the gradient descent method, followed by a diffusion strategy to approximate the global optimal parameters gradually. Mathematically, each agent n performs the following steps:

$$\varphi_n(t+1) = \psi_n(t) - \beta_o(t)\nabla_{\psi_n}\mathcal{L}_n(\psi_n(t)), \tag{8a}$$

$$\psi_n(t+1) = \sum_{m=1}^{N} a_{nm}\varphi_m(t+1). \tag{8b}$$

Here $\beta_o(t)$ is the outer learning rate satisfying (Robbins & Monro, 1951; Tu & Sayed, 2012)

$$\sum_{t=1}^{\infty} \beta_o(t) = \infty, \sum_{t=1}^{\infty} [\beta_o(t)]^2 < \infty.$$

A feasible setting for the outer learning rate $\beta_o(t)$ is

$$\beta_o(t) = \frac{\beta_0}{1+\eta t}, 0 < \eta < 1, 0 < \beta_0 \leq 1,$$

where β_0 denotes the initial outer learning rate and η is the decay coefficient. $\{a_{nm}\}$ are non-negative combination weights satisfying

$$\sum_{m=1}^{N} a_{nm} = 1, a_{nm} = 0 \text{ if } m \notin \mathcal{N}_n.$$

Remark 1. Combination weights $\{a_{nm}\}$ and network topology $\mathcal{G}$jointly determine the efficiency of information diffusion among agents. Practically, common weighting rules include averaging rules, nearest neighbors rules, Laplacian rules, and Metropolis-Hastings rules, among others (Jadbabaie et al., 2003; Olfati-Saber & Murray, 2004; Sayed, 2014; Xiao & Boyd, 2004). Some examples include , and they can even be obtained through some optimization strategies (Takahashi et al., 2010). For simplicity and to validate the robustness of the proposed framework, the main experiments in this paper adopt averaging rules. Their specific mathematical definitions given as follows:

$$a_{nm} = \begin{cases} \frac{1}{|\mathcal{N}_n|}, & m \in \mathcal{N}_n \\ 0, & otherwise \end{cases}$$

Notably, the DVBML framework exhibits robustness to weight selection, and detailed analysis on the impact of different weighting strategies (e.g., Metropolis-Hastings) as well as corresponding performance comparisons are provided in Appendix C.

The outer update Eq. (8) is commonly referred to as the adapt-then-combine (ATC) strategy, which consists of two steps. The first step Eq. (8a) involves local adaptation, where agent n uses its own task set $\mathcal{T}_n$ to update its parameters estimate from $\psi_n(t)$ to an intermediate value $\varphi_n(t+1)$. Then the second step Eq. (8b) is a combination step, where the intermediate estimates $\varphi_n(t+1)$ from the neighboring agent are combined using the weights $\{a_{nm}\}$, resulting in the updated parameter $\psi_n(t+1)$. This strategy leverages distributed optimization theory to aggregate local gradients, ensuring collaborative convergence without central data pooling. While finding a global optimum is intractable due to the non-convex nature of deep neural networks and the bi-level structure of the ELBO, we can establish theoretical guarantees for DVBML converging to a first-order stationary point. Furthermore, we can theoretically quantify the performance improvement brought by the diffusion strategy compared to non-cooperative learning. Detailed theoretical analysis is provided in Section 3.4.

We summarize the proposed distributed variational bayesian meta-learning algorithm in Algorithm 1. Firstly, we randomly initialize the meta-parameters for each agent. Then, each agent performs inner update based on local task set, and then compute the intermediate states by outer update. Finally, by using the diffusion strategy, we combing the intermediate states of all agents and get the updated meta-parameters. In the next subsections, we specifically present a gradient descent-based DVBML algorithm under Gaussian parameterization.

Algorithm 1 Distributed variational Bayesian meta-learning, DVBML.

Require: Dataset $\{\mathcal{D}_{n,i}\}_{n=1}^{N}$, Inner steps K, Learning rate β_{in}, β_o.
  Randomly initialize: $\{\psi_n(0)\}_{n=1}^{N}$.
  for $t = 0, 1, \cdots$ **do**
    for $n = 1, \cdots, N$ **do**
      For each task $\tau_{n,i}$, do inner update:
      $\lambda_{n,i}^{(0)} = \psi_n(t)$,
      for $k = 0, 1, \cdots, K-1$ **do**
        $\lambda_{n,i}^{(k+1)} = \lambda_{n,i}^{(k)} - \beta_{in}\nabla_{\lambda} L(\lambda_{n,i}^{(k)}; \mathcal{D}_{n,i}^{s})$.
      end for
      Compute intermediate states by Eq.(8a):
      $\varphi_n(t+1) = \psi_n(t) - \beta_o \nabla_{\psi_n} \mathcal{L}_n(\psi_n(t))$.
    end for
    for $n = 1, \cdots, N$ **do**
      Update meta parameters by combing intermediate states :
      $\psi_n(t+1) = \sum_{m=1}^{N} a_{nm}\varphi_m(t+1)$.
    end for
    $t = t+1$
  end for

3.2. Gradient-based DVBML algorithm

We will start by providing a detailed explanation of how to perform DVBML based on gradient descent. The prior $p(\phi_i|\theta)$ represents a modeling assumption that offers flexibility in its selection. For simplicity and ease of implementation, we consider fully factorized Gaussian distributions. The posterior $q(\phi_i; \lambda_i)$ is also a flexible function that can well trade off performance and complexity. We consider the models as a neural network. Specifically, let θ denote the neural network parameters, which follows a Gaussian distribution with mean $\mu_\theta \in \boldsymbol{R}^d$ and variance $\sigma_\theta^2 \in \boldsymbol{R}^d$. Since all agents share the same parameter distribution, we omit the subscript n for simplicity. Additionally, we assume $p(\phi_i|\theta)$ to be

$$p(\phi_i|\theta) = \mathcal{N}(\phi_i; \mu_\theta, \sigma_\theta^2). \tag{9}$$

Then $q(\phi_i; \lambda_i)$ is

$$q(\phi_i; \lambda_i^{(K)}) = \mathcal{N}(\phi_i; \mu_{\lambda_i}^{(K)}, \sigma_{\lambda_i}^{2(K)}), \tag{10}$$

where $\lambda_i^{(K)} = \{\mu_{\lambda_i}^{(K)}, \sigma_{\lambda_i}^{2(K)}\}$, $\lambda_i^{(K)}$ is the adapted parameters from the inner update of Eq. (7). The prior $p(\theta)$ should be conjugate to $q(\phi_i; \lambda_i)$, so we let $p(\theta)$ be

$$p(\theta) = \mathcal{N}(\mu; \boldsymbol{0}, \boldsymbol{I}) \prod_{d=1}^{d} \Gamma(\sigma_d^{-2}; a_0, b_0), \tag{11}$$

where a_0, b_0 are the alpha and beta parameters for the gamma distribution.

For numerical stability, we parameterise the standard deviation point-wisely as $\sigma = \exp\rho$ when performing gradient update for the standard deviations of model parameters, i.e., the meta-parameters $\psi = \{\mu_\theta, \exp(\rho_\theta)\}$ is the initial mean and standard deviation of variational parameters $\lambda_i = \{\mu_{\lambda_i}, \exp(\rho_{\lambda_i})\}$.

Now, we consider the inner update. Note that the core of the inner update is the computation of the gradient $\nabla_{\lambda_i} L(\lambda_i; \mathcal{D}_i^s)$, where

$$L(\lambda_i; \mathcal{D}_i^s) = \mathbb{KL}[q(\phi_i; \lambda_i)||p(\phi_i|\theta)] - \mathbb{E}_{q(\phi_i;\lambda_i)}[\log p(\mathcal{D}_i^s|\phi_i)]. \tag{12}$$

For the $\mathbb{KL}$ term in Eq. (12), we can employ the closed-form formula of the $\mathbb{KL}$ divergence between two Gaussian distributions to speed up the training process.

$$\mathbb{KL}[q(\phi_i; \lambda_i)||p(\phi_i|\theta)] = \rho_\theta - \rho_{\lambda_i} + \frac{\exp(2\rho_{\lambda_i}) + (\mu_{\lambda_i} - \mu_\theta)^2}{2\exp(2\rho_\theta)} - \frac{1}{2} \tag{13}$$

Regarding the expectation term $\mathbb{E}_{q(\phi_i;\lambda_i)}[\log p(\mathcal{D}_i^s|\phi_i)]$ in Eq. (12), we utilize Monte Carlo sampling to approximate it. This approach involves

averaging over a set of samples drawn from the approximate posterior distribution, as has been done in most works (Blundell et al., 2015; Gordon et al., 2018; Kingma & Welling, 2013; Ravi & Beatson, 2018). Mathematically, this process can be expressed as follows:

$$\mathbb{E}_{q(\phi_i;\lambda_i)}[\log p(\mathcal{D}_i^s|\phi_i)] \approx \frac{1}{M}\sum_{l=1}^{M}\log p(\mathcal{D}_i^s|\phi_{i,l}), \tag{14}$$

where $\phi_{i,l}$ is sampled from $q(\phi_i;\lambda_i)$, i.e., $\phi_{i,l}\sim q(\phi_i;\lambda_i)$, M is the number of Monte Carlo samples. Note that the log-likelihood $\log p(\mathcal{D}_i^s|\phi_{i,l})$ corresponds to the loss function of the neural network, such as mean square error or cross entropy. We also implement the re-parameterization trick (Kingma & Welling, 2013; Rezende et al., 2014) to enable optimization through Back propagation:

$$\phi_{i,l} = \mu_{\lambda_i} + \delta_l \odot \exp(\rho_{\lambda_i}), \tag{15}$$

where $\delta_l \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$, and $\odot$ is signified an element-wise product. Then we can derive the gradient of the objective function $L(\lambda_i;\mathcal{D}_i^s)$ with respect to λ_i, i.e.,

$$\nabla_{\mu_{\lambda_i}} L(\lambda_i;\mathcal{D}_i^s) = \frac{\partial L(\lambda_i;\mathcal{D}_i^s)}{\partial \phi_{i,l}} + \frac{\partial L(\lambda_i;\mathcal{D}_i^s)}{\partial \mu_{\lambda_i}}, \tag{16a}$$

$$\nabla_{\rho_{\lambda_i}} L(\lambda_i;\mathcal{D}_i^s) = \frac{\partial L(\lambda_i;\mathcal{D}_i^s)}{\partial \phi_{i,l}}\delta_m \odot \exp(\rho_{\lambda_i}) + \frac{\partial L(\lambda_i;\mathcal{D}_i^s)}{\partial \rho_{\lambda_i}}. \tag{16b}$$

Remark 2. From a theoretical perspective, the performance of DVBML improves given more data points in the support set $\mathcal{D}_i^s$. In a standard C-way S-shot classification task, the support set contains $C\times S$ independent samples. Under the i.i.d. assumption, the log-likelihood term in the local ELBO objective (Eq. (12)) decomposes into a sum over all samples: $\log p(\mathcal{D}_i^s|\phi_i) = \sum_{j=1}^{C\times S}\log p(D_{i,j}^s|\phi_i)$. Consequently, the magnitude of the expected data log-likelihood scales linearly with the number of shots S, whereas the KL divergence prior penalty remains invariant to S. As S increases, the data evidence term asymptotically dominates the objective. According to the Bernstein-von Mises theorem (Bochkina, 2023), the variance of the approximate posterior $q(\phi_i;\lambda_i)$ shrinks at a rate of $\mathcal{O}(1/S)$. This mathematical property ensures that increasing the data volume inherently reduces the epistemic uncertainty of the meta-parameters, guaranteeing improved predictive accuracy and tighter, better-calibrated confidence estimates. This will be further verified in the simulations.

For the outer update, approximating the expectation term $\mathbb{E}_{q(\theta;\psi_n)}$ through Monte Carlo sampling is computationally intensive and infeasible. As a result, we use a point estimate for the global parameters θ, indicated by $q(\theta;\psi_n) = \delta_{\theta^*}(\theta)$, where $\delta_y(x) = 1$ if $x = y$, and 0 otherwise. This method has been employed in previous works (Gordon et al., 2018; Ravi & Beatson, 2018; Yoon et al., 2018).

Considering that $\mathcal{T}_n$ would be large in practical application, we can use mini-batch tasks to update the global parameters and thus speed up the training process. The $\mathbb{KL}$ term in the objective $\mathcal{L}_n(\psi_n)$ can be interpreted as a regularizer, encouraging the approximate posterior to remain close to the prior distribution $p(\theta)$. To control the influence of the $\mathbb{KL}$ term within $\mathcal{L}_n$, we introduce a regularization coefficient γ, resulting in the following form (Higgins et al., 2017):

$$\mathcal{L}_n(\psi_n) = \frac{1}{|\mathcal{T}_b|}\sum_{i=1}^{|\mathcal{T}_b|} L(\lambda_{n,i}^{(K)}, \mathcal{D}_{n,i}^q) + \frac{\gamma}{N}\mathbb{KL}[q(\theta;\psi_n)||p(\theta)],$$

where $|\mathcal{T}_b|$ is referred to as the meta-batch size.

Note that $\lambda_{n,i}^{(K)}$ depends on the initialization variational parameters ψ_n through the inner update Eq. (7). This implies that the gradient $\nabla_{\psi_n}\mathcal{L}_n(\psi_n)$ can be computed via backpropagation when optimizing the parameters ψ_n. While the backpropagation step requires computing the Hessian, it can be done efficiently with fast Hessian-vector products, as have been used in previous works (Finn et al., 2017; Kim et al., 2018).

Remark 3. In fact, the proposed gradient-based DVBML algorithm is similar to the distributed sub-gradient optimization algorithm (Nedic & Ozdaglar, 2009) or Consensus-based Distributed Stochastic Gradient Descent (Jiang, 2018). Although the objective function $\mathcal{L}_n(\psi)$ contains an inner update process, as long as L is convex w.r.t. the meta parameters ψ, the gradient-based DVBML can obtain the optimal solution (Nedic, 2020). However, for the setup of complex models such as neural networks, it is generally difficult to ensure that the objective function is convex, so only local optimal or sub-optimal solutions can be obtained. In spite of this, the subsequent simulations will show that all agents can reach consensus and approximate the solution of the centralized algorithm.

3.3. Complexity analysis

In this section, we theoretically analyze the Gradient-based DVBML algorithm from three dimensions: computational complexity, space complexity, and communication complexity, and compare it horizontally with centralized Variational Bayesian Meta-Learning (VBML) and Dif-MAML.

For ease of reference, we recall the notations already established earlier: N is the number of agents, K is the number of inner-loop gradient updates, M is the number of Monte Carlo samples, and $|\mathcal{T}_b|$ is the meta-batch size per agent per outer iteration. The remaining symbols are defined as follows: d is the total dimension of model parameters; $|\mathcal{N}_n|$ is the average number of neighbors per agent; and E is the total number of edges in the communication topology graph.

Computational Complexity. Each iteration of DVBML consists of three parts: inner update (Eq. (7)), outer gradient computation (Eq. (8a)), and diffusion aggregation (Eq. (8b)). In the inner update, each inner step for a single task includes closed-form KL divergence calculation (complexity $O(d)$) and M forward-backward propagations via the reparameterization trick (single propagation complexity $O(d)$). Thus, the computational complexity of K inner updates for one task is $O(KMd)$. In the outer gradient computation, each agent processes $|\mathcal{T}_b|$ tasks and computes meta-gradients via second-order backpropagation, with an overall complexity of $O(|\mathcal{T}_b|KMd)$, which is consistent with the per-iteration gradient computation logic of centralized VBML. In the diffusion aggregation, each agent performs weighted averaging of intermediate parameters from neighboring nodes, with complexity $O(|\mathcal{N}_n|d)$. Under sparse communication topologies, $|\mathcal{N}_n| \ll N$, and this overhead is negligible compared to gradient computation. Therefore, the total per-iteration computational complexity per agent is $O(|\mathcal{T}_b|KMd)$. Compared with the centralized VBML, which processes all tasks on a single node with per-iteration complexity $O(|\mathcal{T}|KMd)$, DVBML distributes the computation across N agents and achieves near-linear wall-clock speedup under ideal conditions. Compared with Dif-MAML, a deterministic point-estimation method with per-forward-pass complexity $O(d)$ that models only the parameter mean, DVBML additionally models the variance, effectively doubling the parameter dimension and incurring roughly twice the per-iteration computational cost of Dif-MAML—a necessary overhead for uncertainty quantification.

Space Complexity. DVBML adopts a fully decentralized architecture wherein each agent stores locally only two categories of data: the mean and variance vectors of meta-parameters, entailing a memory complexity of $O(d)$, and the intermediate parameter vectors received from neighboring nodes, which incur a complexity of $O(|\mathcal{N}_n|d)$. Therefore, the space complexity per agent is $O(d)$, which is in the same order as centralized VBML and Dif-MAML. No global storage node is required, making it naturally suitable for edge deployment scenarios.

Communication Complexity. In each outer update, every agent broadcasts its intermediate parameter vector, comprising both mean and variance and totaling $2d$ scalars, to all neighboring nodes, resulting in a per-iteration network-wide communication volume of $O(E\cdot d)$. Relative to Dif-MAML, which transmits only the parameter means (d scalars) per message, DVBML incurs twice the per-message communication cost. In contrast to federated Bayesian meta-learning with a client-server

architecture, DVBML eliminates the central node bandwidth bottleneck and offers superior scalability in large-scale peer-to-peer networks.

3.4. Theoretical analysis

To address how the multi-agent consensus theoretically guarantees performance improvement, we conceptualize DVBML as a decentralized stochastic non-convex optimization problem. Note that $\mathcal{L}(\psi) = \frac{1}{N}\sum_{n=1}^{N}\mathcal{L}_n(\psi)$ is the global objective. We introduce the following standard assumptions commonly used in multi-agent distributed optimization (Nedic & Ozdaglar, 2009; Nedic et al., 2010).

Assumption 1 (ℓ-smoothness)**.** The local objective function $\mathcal{L}_n(\psi)$ is continuously differentiable and its gradient is ℓ-smooth, i.e., there exists a constant ℓ such that for any ψ_1, ψ_2,

$$||\nabla\mathcal{L}_n(\psi_1) - \nabla\mathcal{L}_n(\psi_2)|| \leq \ell||\psi_1 - \psi_2||.$$

Assumption 2 (Bounded Variance)**.** The stochastic gradient $g_n(\psi)$ computed via Monte Carlo sampling in the outer update is unbiased, and its variance is strictly bounded by a constant κ^2:

$$\mathbb{E}[g_n(\psi)] = \nabla\mathcal{L}_n(\psi), \quad \mathbb{E}[||g_n(\psi) - \nabla\mathcal{L}_n(\psi)||^2] \leq \kappa^2.$$

Assumption 3 (Network Topology.)**.** The weighted adjacent matrix $\mathcal{A} = [a_{nm}]$ is doubly stochastic. Its second largest eigenvalue ϱ satisfies $\varrho < 1$, which guarantees the connectivity and information diffusion efficiency of the network.

Based on these assumptions, we can establish the convergence properties of DVBML and theoretically explain its performance improvement over non-cooperative methods. Let $\bar{\psi}(t) = \frac{1}{N}\sum_{n=1}^{N}\psi_n(t)$ denote the global average of the meta-parameters at iteration t.

Theorem 1 (Convergence and Performance Guarantee.)**.** *Under Assumptions 1–3, by choosing the outer learning rate $\beta_o = \mathcal{O}(\sqrt{\frac{N}{T}})$, the average sequence generated by DVBML after T iterations satisfies*

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[||\nabla\mathcal{L}(\bar{\psi}(t))||^2] \leq \mathcal{O}(\frac{1}{\sqrt{NT}}) + \mathcal{O}(\frac{N}{T}).$$

When the total number of iterations T is sufficiently large (specifically $T \geq N^3$), the first term dominates, and DVBML achieves a linear speedup $\mathcal{O}(\frac{1}{\sqrt{NT}})$ with respect to the number of agents N.

The detailed proof is provided in Appendix D.

Remark 4. Theorem 1 theoretically guarantees the performance improvement of DVBML by demonstrating a linear speedup in variance reduction. This can be derived directly from the convergence bound established in Appendix D. Specifically, the dominant terms governing the convergence rate before substituting the learning rate β_o are the initialization error and the variance term: $\mathcal{O}(\frac{1}{\beta_o T}) + \mathcal{O}(\frac{\beta_o\kappa^2}{N})$.

Non-cooperative Setting ($N = 1$). When agents train in isolation without communication, the optimal learning rate that minimizes the bound is $\beta_o = \mathcal{O}(\frac{1}{\sqrt{T}})$. Substituting this into the variance term yields a convergence rate heavily penalized by local gradient noise, dominated by $\mathcal{O}(\frac{\kappa^2}{\sqrt{T}})$.

DVBML Setting ($N > 1$). By incorporating the diffusion consensus strategy, the multi-agent network allows for a larger optimal learning rate $\beta_o = \mathcal{O}(\sqrt{\frac{N}{T}})$. Substituting this into the variance term $\frac{\beta_o\ell\kappa^2}{N}$ mathematically reduces the noise penalty to $\mathcal{O}(\frac{\kappa^2}{\sqrt{NT}})$. This theoretical comparison proves that the diffusion consensus strategy acts as a global variance reducer. By shrinking the gradient variance κ^2 by a factor of $\sqrt{N}$, DVBML enables agents to converge to a tighter neighborhood of the stationary point compared to isolated training. This rigorously explains the superior generalization accuracy and well-calibrated uncertainty estimation observed in our empirical results.

4. Simulations

We evaluate the performance of the proposed DVBML framework using standard benchmarks for few-shot meta-learning. Our simulation aims to address two primary questions: (1) Can DVBML approach the performance of traditional centralized Bayesian meta-learning in a fully decentralized setting? (2) Compared to existing distributed meta-learning methods, what specific advantages does our algorithm offer, particularly in handling task uncertainty?

To comprehensively answer these questions, we design a multi-faceted comparison that positions DVBML within a complete performance landscape. We primarily compare DVBML against three representative and informative baselines in regression and classification tasks.

- Centralized VBML: This serves as the performance upper bound, where a single agent has access to all data and tasks, executing the standard variational Bayesian meta-learning algorithm. It represents the ideal, non-distributed scenario.
- Non-cooperative VBML: This acts as the performance lower bound, where each agent learns solely from its local data without any communication. It isolates the benefit of collaboration in our framework.
- Dif-MAML: This is the state-of-the-art deterministic distributed meta-learner. It is a derivative approach that extends the classical MAML algorithm to distributed scenarios.

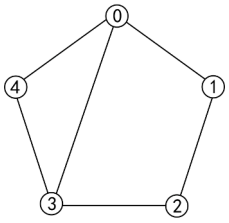
This multi-faceted experimental design allows us to systematically validate DVBML's performance in achieving collaborative consensus, its advantage over isolated learning, and its unique capability in uncertainty-aware learning. Unless otherwise specified, the following experimental setup is used consistently throughout. The data is randomly dispersed across a multi-agent system consisting of five agents. The communication topology and the corresponding diffusion weight matrix $\mathcal{A} = [a_{nm}]$ for this system are illustrated in Fig. 2.

4.1. Regression

We first address a 1D regression problem, where different tasks correspond to different underlying functions. This setup involves a task distribution that is multi-modal (Nguyen et al., 2020). Specifically, half of the functions are sinusoids, and the remaining half are linear. The sinusoid functions are defined as $y = b\sin(wx) + \delta$. Here, the phases parameter w is uniformly sampled from the range $[0, \pi]$, and δ is Gaussian noise with $\delta \sim \mathcal{N}(0, 0.3)$. The amplitude b is divided into five distinct non-overlapping intervals within the range $[0.1, 5.0]$, with each interval assigned to an agent. The lines function is given by $y = cx + d + \delta$, where the intercept d is sampled uniformly from the range $[-3, 3]$. The slope range of $[-3, 3]$ is divided equally among the five agents without overlapping. The input domain x is uniform distributed in the interval $[-5, 5]$, which is consistent for both sinusoidal and linear functions.

The regression model employed here is a neural network similar to the one used in Kayaalp et al. (2022), which consists of two hidden layers of size 40 utilizing ReLU activation functions. The loss function is the mean-squared error (MSE) between the prediction and the true value.

To qualitatively evaluate the performance of different algorithms, we randomly sample four different tasks and perform regression tests with all algorithms. The results are presented in Fig. (3), where the red points are support set, and the solid lines indicate ground truth of the sampled function. The shaded area is the prediction made by variational Bayesian meta-learning algorithms with $\pm 1\times$ standard deviation. The results show that Dif-MAML can only output single-value prediction at each input, because it optimizes point estimates of parameters without modeling their distributions. In contrast, DVBML is able to predict the probability distribution (the mean and variance) of the output, since DVBML models parameters in probability space. On the other hand, DVBML can effectively reason about distribution of the underlying functions that generates the support data points. Furthermore, it is observed that the



$$\mathcal{A} = \begin{bmatrix} 1/4 & 1/4 & 0 & 1/4 & 1/4 \\ 1/3 & 1/3 & 1/3 & 0 & 0 \\ 0 & 1/3 & 1/3 & 1/3 & 0 \\ 1/4 & 0 & 1/4 & 1/4 & 1/4 \\ 1/3 & 0 & 0 & 1/3 & 1/3 \end{bmatrix}$$

Fig. 2. Graph underlying the network with corresponding weight matrix.

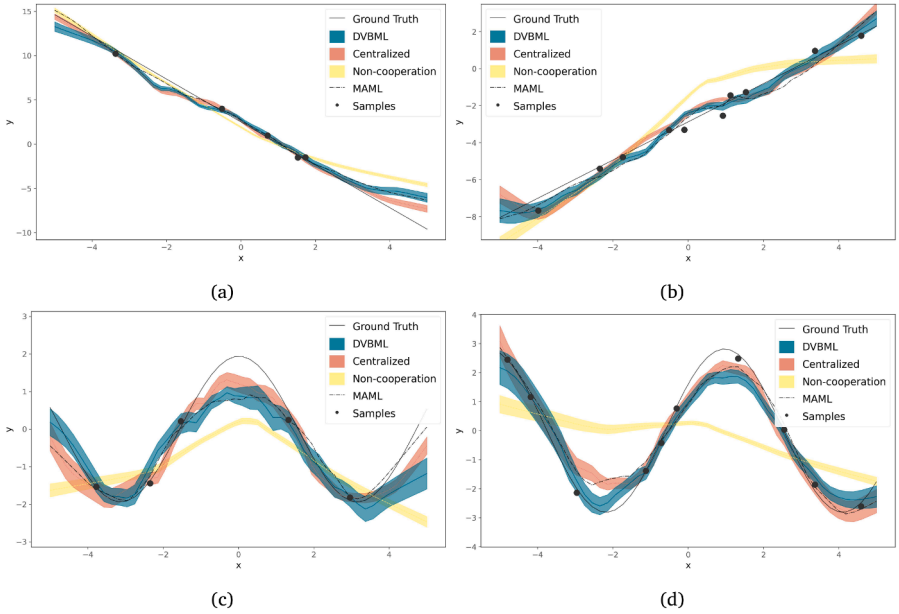


Fig. 3. Qualitative results on multi-modal regression task: (a) linear data for 5-shot, (b) linear data for 10-shot, (c) sinusoidal data for 5-shot, and (d) sinusoidal data for 10-shot.The shaded area represents ± 1 standard deviation of the predictive distribution (estimated based on the variance of the variational posterior $q(\phi_i; \lambda_i)$).

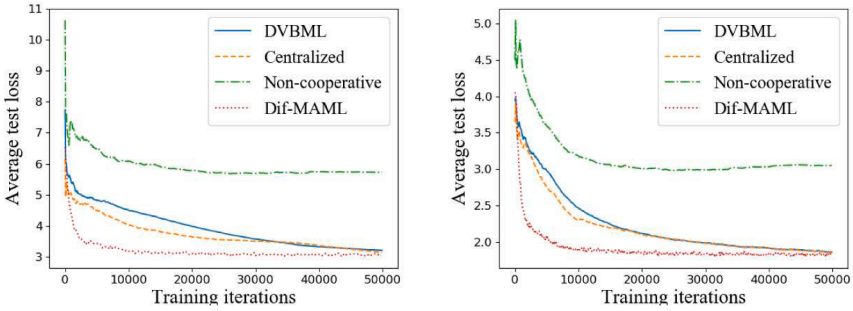
variance of the learned function in DVBML is lower than that of the non-cooperative VBML, and is more closely to the centralized VBML. This is because in the DVBML algorithm, the agent shares parameters with the neighbors, so that it can learn the global optimal parameter distribution, as obtained by the centralized algorithm.

In Fig. 4, we present the average MSE performance across 1000 test tasks. These test tasks consist of both sinusoidal and linear functions. Unlike the training task for each agent, the amplitude of the sinusoids is in the range of $[0.1, 5.0]$, while the slope of the lines fall within the range of $[-3, 3]$. It is clear that for both 5-shot and 10-shot regression tasks, the non-cooperative algorithm gets stuck at a local optimum, leading to a significant bias due to the agent only utilizes local data and information in every update step. In contrast, DVBML improves performance gradually with the information diffused from the whole multi-agent systems. As observed, DVBML tends to approach the centralized solution over time, which is not the case for the non-cooperative algorithm. This reveals that sharing model parameters with neighbors facilitates all agents to approximate the centralized solution.

Uncertainty Calibration. To quantitatively evaluate the quality of uncertainty estimates, we employ reliability diagrams along with the Expected Calibration Error (ECE) and Maximum Calibration Error (MCE) (Jeon et al., 2024; Nguyen et al., 2020; Song et al., 2019), with results shown in Fig. 5. A model is perfectly calibrated when its predicted probability equals to the actual probability, resulting in a curve that is well-aligned with the diagonal $y = x$. Our primary findings are twofold. First, DVBML effectively approximates the performance of the centralized VBML and significantly outperforms the non-cooperative baseline. This is attributed to its diffusion-based parameter fusion, which enables consensus learning across agents by aggregating diverse task information. Second, in contrast to the deterministic Dif-MAML—whose reliability diagram degenerates into a horizontal line due to the deterministic nature—DVBML provides well-calibrated uncertainty quantification. Its reliability curve is closer to the diagonal than that of Dif-MAML, while it also achieves lower ECE and MCE values in comparison. This advantage stems directly from its probabilistic modeling of parameters via variational inference, which inherently captures task uncertainty. A detailed examination of Fig. 5 reveals an expected trend: calibration improves for all methods from the 5-shot to the 10-shot setting, evidenced by lower ECE/MCE and reliability curves shifting closer to the diagonal. Deterministic methods (e.g., Dif-MAML) primarily exploit additional data to refine point estimates without explicitly improving confidence quality. In contrast, DVBML naturally translates the larger volume of data into posterior variance shrinkage. This empirically corroborates our theoretical analysis: increasing the number of data points drives the asymptotic shrinkage of the posterior variance, thereby inherently reducing task-specific epistemic uncertainty and yielding strictly better-calibrated predictions.

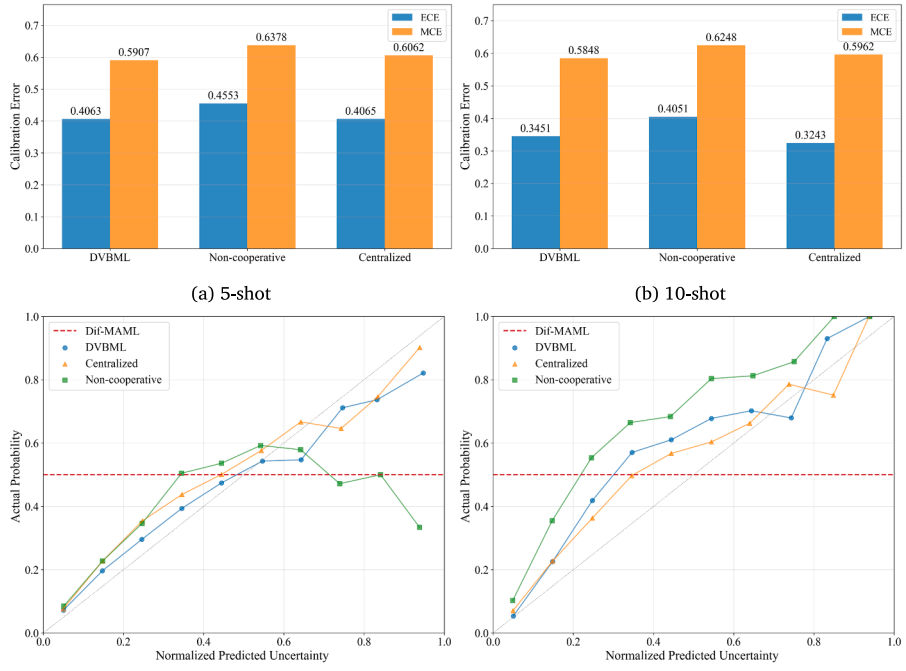
4.2. Classification

To evaluate the performance of proposed method on a more complex model, we conduct evaluations on two widely used few-shot classification datasets: Omniglot (Lake et al., 2011) and MiniImageNet (Vinyals et al., 2016). Our experiments are designed based on the C-way S-shot learning task, with each individual task contains C classes. Within each task, the support set (used for inner update) consisted of S samples per



(a) 5-shot (b) 10-shot

Fig. 4. Average test loss on regression task: (a) 5-shot, (b) 10-shot. The "DVBML" is the average loss of all agents. The "Non-cooperative" is one randomly chosen agent. The following figures are the same.



(a) 5-shot (b) 10-shot

(c) 5-shot (d) 10-shot

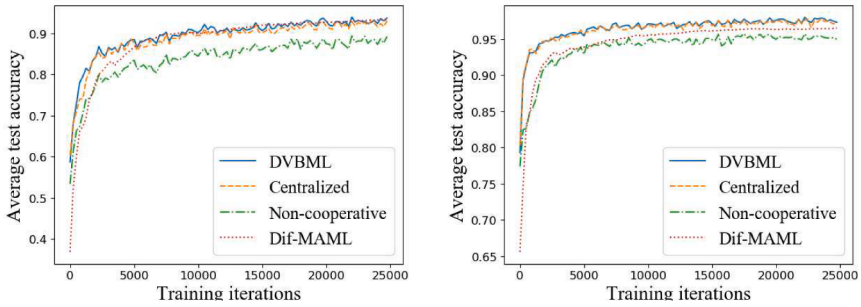
Fig. 5. Quantitative results on multimodal data —— half of the tasks are generated from sinusoidal functions, and the other half are from linear functions: (a) and (b) are ECE and MCE of various metalearning methods averaged over 1000 tasks, (c) and (d) are Reliability diagram of different methods..

class, while the query set (employed for outer update) contained 15 examples. The communication topology and the corresponding diffusion weight matrix $\mathcal{A} = [a_{nm}]$ for this simulation are illustrated in Fig. 2.

The Omniglot dataset involves 1623 distinct characters, each having 20 instances. These images are resized to dimensions of 28×28 pixels. The number of characters for meta-training, meta-validation and meta-test are 1100, 100, and 423, respectively. The MiniImageNet dataset consists of total 100 different classes (64 for meta-training, 16 for meta-validation, and 20 for meta-testing Ravi & Larochelle, 2017). Each class contains 600 color images, all of which are of size 84×84 pixels. The training dataset for both datasets is randomly partitioned into five balanced, non-overlapping subsets, with each agent receiving approximately 20% of the data in standard experiments. For imbalanced scenarios, specific agents are allocated fewer classes/data to test robustness. Each part was then allocated to a single agent. Consequently, each agent has about 220 characters for the Omniglot dataset and 13 classes for MiniImageNet dataset.

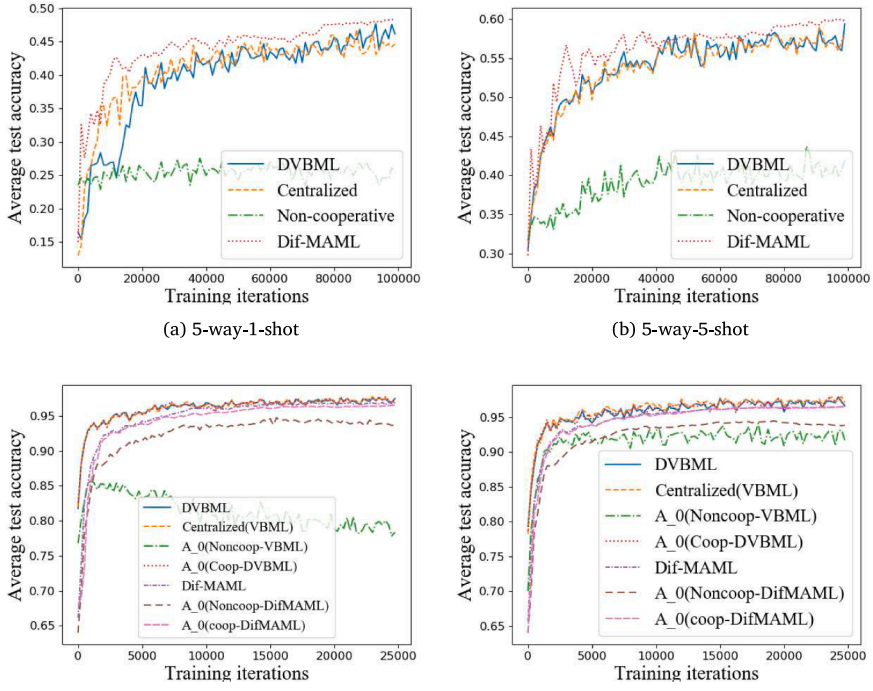
Both for the Omniglot and MiniImageNet datasets, we employ a widely-used network architecture that has been previously used in various studies (Kayaalp et al., 2022; Ravi & Beatson, 2018). The model consists of four modules with 3-by-3 stride convolutions and 64 filters, followed by batch normalization and a ReLU activation. For the MiniImageNet dataset, we use a max-pooling layer instead of stride convolution in each module and use 32 filters per layer to reduce overfitting (Ravi & Beatson, 2018; Ravi & Larochelle, 2017). In both cases, the loss function is the cross-entropy error, a common choice for classification tasks. Please see Appendix B for more simulation details.

Predictive Performance. To ensure a fair comparison, we evaluate DVBML against deterministic Dif-MAML and other baselines using identical network architectures and data assignments. We test the average test accuracy across 1000 sampled tasks, with assessments performed every 1000th iteration. As illustrated in Figs. 6 and 7, the training performance of DVBML gradually approximates that of the centralized VBML. Conversely, the non-cooperative approach fails to converge to the centralized solution—consistent with our regression results—because isolated agents lack interaction and suffer from limited local data diversity. While centralized algorithms inherently maintain a slight performance upper bound over distributed ones, Table 1 demonstrates that



(a) 5-way-1-shot (b) 5-way-5-shot

Fig. 6. Average test accuracy on Omniglot for (a) 5-way-1-shot and (b) 5-way-5-shot.



(a) 5-way-1-shot (b) 5-way-5-shot

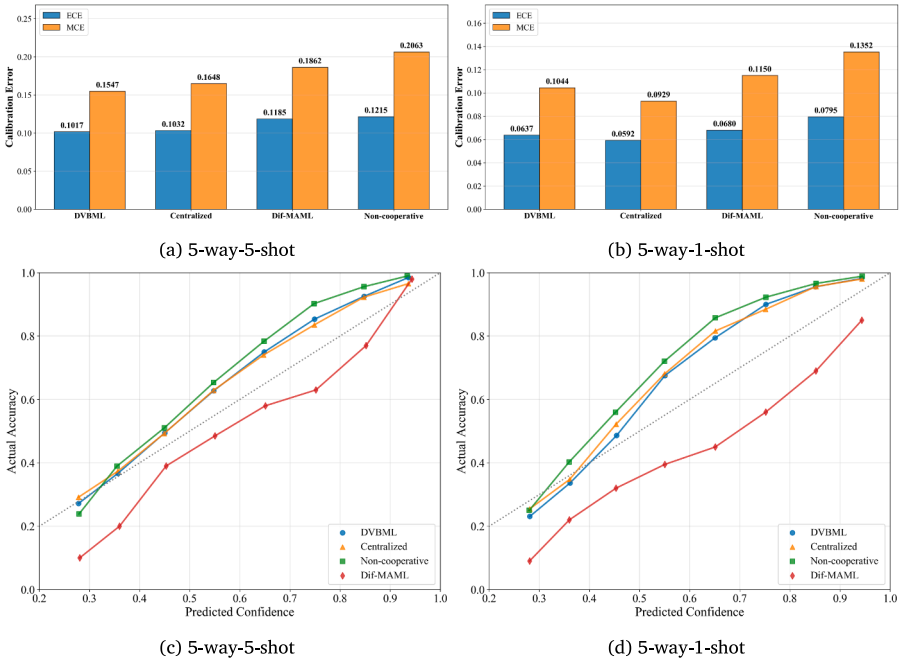
(c) 5-way-5-shot (d) 5-way-5-shot

Fig. 7. Average test accuracy on MiniImagenet for (a) 5-way-1-shot and (b) 5-way-5-shot. (c) and (d) are results on Omniglot for imbalanced settings.

DVBML ultimately achieves mean final accuracies highly comparable to the state-of-the-art deterministic Dif-MAML. Crucially, the classification accuracy variance of DVBML is strictly smaller than that of Dif-MAML. This indicates that explicitly modeling parameter distributions empowers DVBML with more stable reasoning and a distinct advantage in predictive reliability.

Uncertainty Calibration. Beyond predictive accuracy, we evaluate the quality of explicit uncertainty quantification on Omniglot and MiniImageNet (Figs. 8 and 9). Through the diffusion consensus strategy (Eq. (8b)), DVBML closely matches the calibration profile of centralized VBML and substantially surpasses the non-cooperative baseline. Furthermore, DVBML significantly outperforms the deterministic Dif-MAML, yielding much lower ECE/MCE metrics and reliability curves that adhere closely to the ideal diagonal. Further analysis reveals two consistent patterns that validate our core Bayesian design: First, calibration strictly improves from the 1-shot to the 5-shot scenario across both datasets, which directly reflects the explicit $\mathcal{O}(1/S)$ posterior variance shrinkage mechanism derived in our theoretical formulation. Second, while the overall calibration on the more complex MiniImageNet dataset is naturally inferior to Omniglot due to greater aleatoric uncertainty in natural images, DVBML exhibits remarkable robustness. Despite a modest mean accuracy trade-off compared to Dif-MAML on this dataset, DVBML suffers the smallest calibration degradation under high-uncertainty conditions. This confirms that integrating distributed collaborative learning with explicit variational modeling successfully generates trustworthy confidence estimates, which is paramount for safety-critical multi-agent applications.

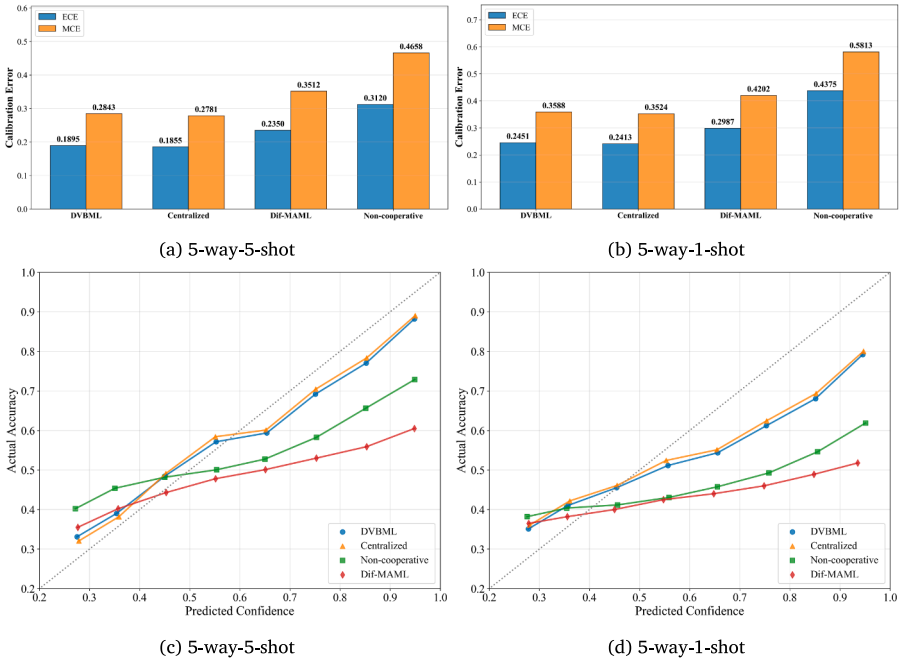
Imbalanced data case. We evaluate the algorithms on imbalance data settings, where some agents only have access to very little data. In one case, we allocate 20 classes of the Omniglot training set to one specific agent (labeled as A_0), while the remaining data is divided equally among the other agents, with each agent having 216 categories. In the other case, we assign A_0 with 50 categories data from Omniglot, each category containing only half the original data (10 images per category). The results are presented in Fig. 7(c) and (d). It can be seen that agent A_0 can approach the centralized solution and outperforms the non-cooperative one even when having limited data sources. The DVBML approach effectively maintains a balance between the



(a) 5-way-5-shot (b) 5-way-1-shot

(c) 5-way-5-shot (d) 5-way-1-shot

Fig. 8. Uncertainty Evaluation of Meta-Learning Methods on Omniglot: Reliability Diagrams, ECE and MCE (5-way 1-shot/5-shot, 1000 unseen Tasks).



(a) 5-way-5-shot (b) 5-way-1-shot

(c) 5-way-5-shot (d) 5-way-1-shot

Fig. 9. Uncertainty Evaluation of Meta-Learning Methods on MiniImageNet: Reliability Diagrams, ECE and MCE (5-way 1-shot/5-shot, 1000 unseen Tasks).

information from local data and that shared by neighboring agents. The results reveal that DVBML can also be stably applied in unbalanced data settings which is commonly found in real-world applications.

4.3. Complexity and ablation studies

4.3.1. Empirical complexity validation

To verify the conclusions of the theoretical complexity analysis, we conduct complexity comparison experiments on the Omniglot 5-way-1-shot task. To ensure fair comparison, all algorithms adopt a unified convergence termination criterion. All experiments are conducted on a single NVIDIA RTX 3090 GPU with PyTorch 1.8.1 and CUDA 11.1. We select four core metrics for comparison: average per-iteration time (corresponding to computational complexity), peak GPU memory usage (corresponding to space complexity), number of iterations to convergence, and total training wall-clock time (corresponding to end-to-end efficiency). The results are shown in Table 2. The experimental results are consistent with the theoretical analysis. As shown in Table 2, because DVBML models both the mean and variance to quantify uncertainty, its total computational cost (2.93 s) and memory footprint (51.56 MB) on a single GPU are roughly twice that of the deterministic Dif-MAML (1.65 s, 28.45 MB). Crucially, the total computational time of DVBML (2.93 s) matches the Centralized VBML (2.92 s). However, it is vital to note that this represents the sequential sum of all 5 agents

Table 1
Few-shot classification accuracies (%) with 95% confidence intervals on Omniglot and MiniImageNet.

Algorithms	Omniglot		MiniImageNet	
	5-way 5-shot	5-way 1-shot	5-way 5-shot	5-way 1-shot
DVBML	98.15±0.42	93.29±0.61	59.75±0.86	45.25±0.95
Centralized VBML	98.24±0.38	94.02±0.55	60.25±0.94	44.86±0.88
Non-cooperative VBML	93.58±0.68	88.25±0.96	40.36±1.86	25.65±2.32
Dif-MAML	97.01±0.89	94.25±1.48	60.58±1.28	45.45±1.68

Table 2
Complexity comparison of different methods.

Metric	Centralized VBML	Dif-MAML	DVBML(ours)
Per-iteration time (s)	2.92	1.65	2.93
Peak GPU memory (MB)	51.49	28.45	51.56
Iterations to convergence	19,824	20,138	19,993
Total training time (min)	964.8	553.8	976.3

Table 3
Ablation results on Omniglot (5-way-5-shot).

Methods	DCS	VB	Classification Accuracy (%)	ECE
Non-cooperative MAML	×	×	94.12 ± 1.12	0.1312
Non-cooperative VBML	×	✓	93.58 ± 0.68	0.1285
Dif-MAML	✓	×	97.01 ± 0.89	0.1185
DVBML (ours)	✓	✓	98.15 ± 0.42	0.1017
Centralized MAML	×	×	98.05 ± 1.41	0.1265
Centralized VBML	×	✓	98.24 ± 0.38	0.1032

simulated on a single hardware node. In a true decentralized multi-agent deployment, DVBML distributes this workload across $N = 5$ independent agents. Consequently, the actual per-node computational time for DVBML would be strictly reduced to approximately $2.93/5 \approx 0.59$ seconds per iteration. This empirical finding validates the theoretical claim in Section 3.3: DVBML achieves a near-linear wall-clock speedup proportional to the network size N, effectively eliminating the computational bottleneck of centralized data processing while preserving Bayesian uncertainty awareness.

4.3.2. Component ablation analysis

To analyze the individual contributions and synergistic effects of the two core components of DVBML, namely the diffusion consensus strategy and variational Bayesian probabilistic modeling, this section conducts a controlled ablation study under the 5-way-5-shot setting on two standard benchmarks, Omniglot and MiniImageNet.

Control variables are defined for the two core modules. The diffusion consensus strategy(DCS) controls whether the adapt-then-combine consensus mechanism among multi-agents is enabled; when disabled, each agent trains independently based only on local data. The variational Bayesian(VB) probabilistic modeling controls whether meta-parameters are modeled as probability distributions; when disabled, the model adopts deterministic point estimation. Based on this, we construct six algorithm variants in total, including four distributed settings and two centralized baselines. All methods share identical network architecture, data partition. The results are shown in Tables 3 and 4. The two sets of ablation experiments exhibit consistent patterns across both datasets, revealing the respective roles of DCS and VB modeling, as well as their synergistic effect.

Role of DCS (Enabling Global Consensus): Integrating DCS is critical for achieving consensus in distributed scenarios. By aggregating multi-agent parameters to compensate for local data scarcity, enabling DCS significantly improves both classification accuracy and ECE compared to the non-cooperative baselines, successfully driving the distributed model to approach the theoretical performance bound of the centralized model.

Table 4
Ablation results on MiniImageNet (5-way-5-shot).

Methods	DCS	VB	Classification Accuracy (%)	ECE
Non-cooperative MAML	×	×	42.65 ± 1.95	0.3365
Non-cooperative VBML	×	✓	40.36 ± 1.86	0.3120
Dif-MAML	✓	×	60.58 ± 1.28	0.2350
DVBML (ours)	✓	✓	59.75 ± 0.86	0.1895
Centralized MAML	×	×	61.02 ± 1.02	0.2412
Centralized VBML	×	✓	60.25 ± 0.94	0.1855

Role of VB (Enabling Uncertainty Quantification): VB modeling is essential for generating reliable confidence estimates. When integrated with DCS, VB substantially reduces ECE compared to the deterministic distributed baseline (Dif-MAML), while maintaining highly competitive classification accuracy. Notably, VB yields a much smaller calibration improvement in the non-cooperative setting. This contrast indicates that the robust global prior constructed via DCS collaboration fundamentally amplifies the advantages of Bayesian modeling.

Synergistic Effect: The combination of DCS and VB achieves performance highly comparable to Centralized VBML in fully decentralized scenarios. Crucially, the calibration improvement achieved by combining both components significantly exceeds the sum of their individual contributions. This confirms that synchronously diffusing Bayesian parameters across agents effectively enhances the reliability of task posterior inference.

Ultimately, these results confirm the effectiveness of the proposed DVBML algorithm, striking an balance between collaborative consensus and rigorous uncertainty quantification.

5. Conclusions and future work

This paper establishes the Distributed Variational Bayesian Meta-Learning (DVBML) framework, enabling collaborative and uncertainty-aware few-shot learning across decentralized multi-agent systems. By synthesizing variational inference with a diffusion-based consensus strategy, the proposed gradient-diffusion hybrid algorithm explicitly models and propagates task uncertainty. Mathematically, we established rigorous theoretical guarantees, proving that DVBML ensures convergence and achieves a linear speedup in variance reduction compared to non-cooperative learning. Extensive empirical evaluations across regression and classification benchmarks corroborate these theoretical findings, demonstrating that DVBML matches centralized Bayesian performance and delivers highly reliable, well-calibrated uncertainty estimates at a controllable computational cost, significantly outperforming deterministic distributed baselines even in imbalanced settings.

Future research can be extended in several directions. First, developing asynchronous communication variants could mitigate idle-time synchronization overheads in highly heterogeneous networks. Second, extending the analysis to dynamic topologies with time-varying communication links would evaluate the algorithm's resilience in unstable edge environments. Finally, exploring non-Gaussian variational posterior approximations while maintaining computational tractability remains an important avenue to further enhance task inference accuracy.

CRediT authorship contribution statement

Xianyang Zhang: Writing – original draft, Validation, Methodology, Conceptualization; **Weijie Kang:** Supervision, Methodology; **Ying Li:** Validation, Formal analysis; **Bing He:** Writing – review & editing, Project administration; **Honglan Huang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. The derivation of the problem (1)

Given observed task set $\mathcal{T}$, the meta-parameter θ can be estimated by integrating the parameters over the specific tasks to obtain the marginal likelihood of the data. Specifically, the log-likelihood of the observed data is

$$\log p(\mathcal{D}) = \log \prod_{i=1}^{|\mathcal{T}|} p(\mathcal{D}_i) = \log \int p(\theta) \left[\prod_{i=1}^{|\mathcal{T}|} \int p(\mathcal{D}_i|\phi_i) p(\phi_i|\theta) d\phi_i \right] d\theta, \tag{A.1}$$

prior of the meta parameters θ, and $p(\phi_i|\theta)$ is the conditional prior for the task-specific parameters ϕ_i. Directly maximizing the likelihood $\log p(\mathcal{D})$ is difficult because of the double integration involved. Therefore, according to the principle of variational inference, the likelihood can be approximated by maximizing the variational evidence lower bound.

By using $q(\theta;\psi)$ to approximate $p(\theta)$, we have

$$\begin{aligned}
&\log \int p(\theta) \left[\prod_{i=1}^{|\mathcal{T}|} \int p(\mathcal{D}_i|\phi_i) p(\phi_i|\theta) d\phi_i \right] d\theta \\
&\geq \mathbb{E}_{q(\theta;\psi)} \left[\log \left(\prod_{i=1}^{|\mathcal{T}|} \int p(\mathcal{D}_i|\phi_i) p(\phi_i|\theta) d\phi_i \right) \right] - \mathbb{KL}[q(\theta;\psi)||p(\theta)] \\
&= \mathbb{E}_{q(\theta;\psi)} \left[\sum_{i=1}^{|\mathcal{T}|} \log \left(\int p(\mathcal{D}_i|\phi_i) p(\phi_i|\theta) d\phi_i \right) \right] - \mathbb{KL}[q(\theta;\psi)||p(\theta)],
\end{aligned} \tag{A.2}$$

where $\mathbb{KL}$ is the Kullback-Leibler divergence between two distributions,

$$\mathbb{KL}[q(\theta;\psi)||p(\theta)] = \int q(\theta;\psi) \log \frac{q(\theta;\psi)}{p(\theta)} d\theta \tag{A.3}$$

Furthermore, using $q(\phi_i;\lambda_i)$ to approximate the distribution of task-specific parameters ϕ_i, we obtain

$$\begin{aligned}
&\mathbb{E}_{q(\theta;\psi)} \left[\sum_{i=1}^{|\mathcal{T}|} \log \left(\int p(\mathcal{D}_i|\phi_i) p(\phi_i|\theta) d\phi_i \right) \right] - \mathbb{KL}[q(\theta;\psi)||p(\theta)] \\
&\geq \mathbb{E}_{q(\theta;\psi)} \left[\sum_{i=1}^{|\mathcal{T}|} \mathbb{E}_{q(\phi_i;\lambda_i)} [\log p(\mathcal{D}_i|\phi_i)] - \mathbb{KL}[q(\phi_i;\lambda_i)||p(\phi_i|\theta)] \right] \\
&- \mathbb{KL}[q(\theta;\psi)||p(\theta)] \\
&\triangleq \text{ELBO}.
\end{aligned} \tag{A.4}$$

By define the objective as $\mathcal{L}(\psi) = -\text{ELBO}$, we obtain the description as shown in the problem (1):

$$\begin{aligned}
\arg\max \text{ELBO} \equiv \arg\min_{\psi} \mathcal{L}(\psi) = &\mathbb{E}_{q(\theta;\psi)} \left[\sum_{i=1}^{|\mathcal{T}|} -\mathbb{E}_{q(\phi_i;\lambda_i)} [\log p(\mathcal{D}_i|\phi_i)] \right. \\
&\left. + \mathbb{KL}[q(\phi_i;\lambda_i)||p(\phi_i|\theta)] \right] + \mathbb{KL}[q(\theta;\psi)||p(\theta)].
\end{aligned} \tag{A.5}$$

Appendix B. Simulation settings for Gradient-based DVBML

Regression. For both 5-shot and 10-shot regression, we perform $K = 8$ times gradient descent in the inner update with inner learning rate $\beta_{in} = 0.001$. The outer update optimization is done via Adam optimizer (Kingma & Ba, 2014) with learning rate $\beta_o = 0.0015$. For each iteration, we randomly sample 2 task to update meta parameters, i.e., $|\mathcal{T}_b| = 2$. The data points in query set $\mathcal{D}_i^q$ is 15 for each task. We randomly sample 2 models from $q(\phi_{n,i};\lambda_{n,i})$ for each inner update, i.e., $M = 2$. The regularization coefficient $\gamma = 10^{-6}$. The hyper-prior $a_0 = 1, b_0 = 0.01$.

Classification. Some hyper-parameters for classification task are presented in Table B.1.

Table B.1
Hyper-parameters for Omniglot and MiniImageNet.

Data set	Omniglot		MiniImageNet	
C-way S-shot (C,S)	(5,1)	(5,5)	(5,1)	(5,5)
Inner learning rate β_{in}	0.08	0.1	0.05	0.15
Training shots	1	5	1	5
Test shots	15	15	10	10
Inner updates K	10	10	5	5
Outer learning rate β_o	0.0012	0.001	0.005	0.004
Monte Carlo samples M	4	2	2	2
Meta-batch size $\lvert\mathcal{T}_b\rvert$	1	1	2	1
Eval. inner updates K	10	10	10	10
Regularization coefficient γ	10^{-6}	10^{-6}	10^{-6}	10^{-6}

Appendix C. Comparison experiments of different weighting strategies and result analysis

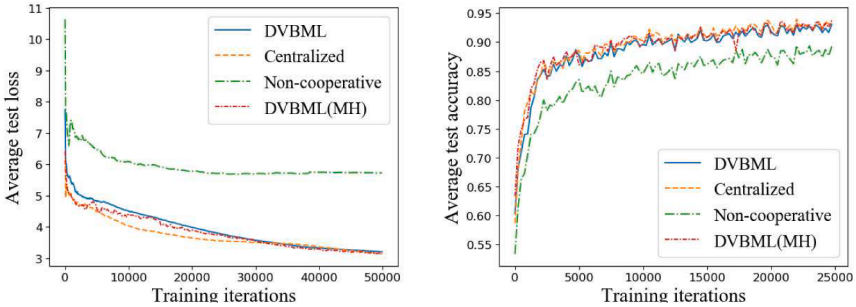
To comparatively analyze the impact of different weighting strategies, we conducted additional simulations using the Metropolis-Hastings weighting strategy alongside the existing average weighting approach. For any pair of agents n and m in the multi-agent systems, the Metropolis-Hastings weight is defined as:

$$\begin{aligned}
a_{nm} &= \frac{1}{\max(|\mathcal{N}_n|, |\mathcal{N}_m|)} \quad \text{for } m \in \mathcal{N}_n \setminus \{n\}, \\
a_{nn} &= 1 - \sum_{m \in \mathcal{N}_n \setminus \{n\}} a_{nm}.
\end{aligned}$$

All comparative experiments were implemented on the network topology illustrated in Fig. 2. For this specific topology (an undirected graph with 5 agents), the corresponding weight matrix of the Metropolis-Hastings strategy is given as follows:

$$\mathcal{A} = \begin{bmatrix} 0 & 1/3 & 0 & 1/3 & 1/3 \\ 1/3 & 1/6 & 1/2 & 0 & 0 \\ 0 & 1/2 & 1/6 & 1/3 & 0 \\ 1/3 & 0 & 1/3 & 0 & 1/3 \\ 1/3 & 0 & 0 & 1/3 & 1/3 \end{bmatrix}$$

The experiments systematically evaluated the performance of these two weighting strategies across both 5-shot regression and Omniglot 5-way-1-shot classification tasks. To establish comprehensive performance benchmarks, we included both centralized and non-cooperative methods as reference baselines, enabling a thorough analysis of how different weighting strategies influence the distributed variational Bayesian meta-learning framework. The experimental results are shown in Fig. C.1. The findings demonstrate that the Metropolis-Hastings weighting strategy exhibits faster convergence during the early training phase, effectively accelerating information fusion among agents while also showing superior training stability. However, after sufficient training iterations, both weighting strategies eventually converge to comparable performance levels, with both significantly outperforming the non-cooperative method and approaching the performance of the centralized method. This discovery validates the robustness of the DVBML framework to different weighting selections, indicating that the simpler average weighting rules already satisfies basic performance requirements, while leaving more complex adaptive weighting optimization for further exploration in future research.



(a) regression 5shot (b) classification 5-way-1-shot

Fig. C.1. Performance comparison of different weighting strategies on 5-shot regression and Omniglot 5-way-1-shot classification tasks.

Appendix D. Proof of Theorem 1

In this section, we provide the detailed proof for the convergence of DVBML. The update rule of DVBML (Eqs. (8a) and (8b)) can be compactly written in a matrix form. We first define the average parameter $\bar{\psi}(t) = \frac{1}{N}\sum_{n=1}^{N}\psi_n(t)$.

Multiplying the update rule by $\frac{1}{N}\sum_{n=1}^{N}$ on both sides, and using the doubly stochastic property of $\mathcal{A}$, the network average updates as follows:

$$\bar{\psi}(t+1) = \bar{\psi}(t) - \frac{\beta_o}{N}\sum_{n=1}^{N} g_n(\psi_n(t)).$$

By the ℓ-smoothness of $\mathcal{L}$ (Assumption 1) and the descent lemma, we have

$$\mathcal{L}(\bar{\psi}(t+1)) \le \mathcal{L}(\bar{\psi}(t)) + \langle \nabla\mathcal{L}(\bar{\psi}(t)), \bar{\psi}(t+1) - \bar{\psi}(t)\rangle + \frac{\ell}{2}||\bar{\psi}(t+1) - \bar{\psi}(t)||^2.$$

Taking the expectation on both sides and substituting the update rule of $\bar{\psi}(t+1)$, we obtain

$$\mathbb{E}[\mathcal{L}(\bar{\psi}(t+1))] \le \mathbb{E}[\mathcal{L}(\bar{\psi}(t))] - \beta_o\mathbb{E}[\langle \nabla\mathcal{L}(\bar{\psi}(t)), \frac{1}{N}\sum_{n=1}^{N} g_n(\psi_n(t))\rangle] + \frac{\beta_o^2\ell}{2}\mathbb{E}[||\frac{1}{N}\sum_{n=1}^{N} g_n(\psi_n(t))||^2].$$

Using the identity $-\langle a, b\rangle = \frac{1}{2}(||a-b||^2 - ||a||^2 - ||b||^2)$ and Assumption 2, we can bound the terms. Specifically, the gradient variance is reduced by a factor of N, i.e.,

$$\mathbb{E}[||\frac{1}{N}\sum_{n=1}^{N} g_n(\psi_n(t))||^2] \le \frac{\kappa^2}{N} + \mathbb{E}[||\frac{1}{N}\sum_{n=1}^{N}\nabla\mathcal{L}_n(\psi_n(t))||^2]$$

Substituting this back into the descent inequality yields

$$\mathbb{E}[\mathcal{L}(\bar{\psi}(t+1))] \le \mathbb{E}[\mathcal{L}(\bar{\psi}(t))] - \frac{\beta_o}{2}\mathbb{E}[||\nabla\mathcal{L}(\bar{\psi}(t))||^2] + \frac{\beta_o\ell^2}{2N}\sum_{n=1}^{N}\mathbb{E}[||\psi_n(t) - \bar{\psi}(t)||^2] + \frac{\beta_o^2\ell\kappa^2}{2N}.$$

Here, the term $\frac{1}{N}\sum_{n=1}^{N}\mathbb{E}[||\psi_n(t) - \bar{\psi}(t)||^2]$ represents the **consensus error** among agents. According to the spectral graph theory (Assumption 3), the consensus error is bounded by the network topology and the gradient variance, i.e.,

$$\frac{1}{N}\sum_{n=1}^{N}\mathbb{E}[||\psi_n(t) - \bar{\psi}(t)||^2] \le \frac{\beta_o^2\kappa^2}{(1-\varrho)^2}$$

Summing the descent inequality over $t = 1$ to T, and dividing by $T\beta_o/2$, we get the final bound for the average gradient norm:

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[||\nabla\mathcal{L}(\bar{\psi}(t))||^2] \le \underbrace{\frac{2(\mathcal{L}(\bar{\psi}(1)) - \mathcal{L}^*)}{\beta_o T}}_{\text{Initialization Error}} + \underbrace{\frac{\beta_o\ell\kappa^2}{N}}_{\text{Variance Term}} + \underbrace{\frac{\beta_o^2\ell^2\kappa^2}{(1-\varrho)^2}}_{\text{Consensus Error}},$$

where $\mathcal{L}^*$ is the global minimum of the objective.

By choosing the outer learning rate $\beta_o = \sqrt{\frac{N}{T}}$, the inequality simplifies to

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[||\nabla\mathcal{L}(\bar{\psi}(t))||^2] \le \frac{2(\mathcal{L}(\bar{\psi}(1)) - \mathcal{L}^*)}{\sqrt{NT}} + \frac{\ell\kappa^2}{\sqrt{NT}} + \frac{N\ell^2\kappa^2}{T(1-\varrho)^2}$$

This gives the final convergence bound

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[||\nabla\mathcal{L}(\bar{\psi}(t))||^2] \le \mathcal{O}(\frac{1}{\sqrt{NT}}) + \mathcal{O}(\frac{N}{T}).$$

This completes the proof. As T becomes sufficiently large, the $\mathcal{O}(\frac{N}{T})$ term approaches zero faster than the $\mathcal{O}(\frac{1}{\sqrt{NT}})$ term. Therefore, the dominant term of the convergence rate is bounded by $\mathcal{O}(\frac{1}{\sqrt{NT}})$.

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